

What does 'good' look like?

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

```
SELECT
column_name,
data_type
FROM `inner-orb-475507-
i2.Target.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers';
```

Query results Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	column_name	data_type			
1	customer_id	STRING			
2	customer_unique_id	STRING			
3	customer_zip_code_prefix	INT64			
4	customer_city	STRING			
5	customer_state	STRING			

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- Insights

Most of the columns in customer table are string.

2. Get the time range between which the orders were placed.

```
select
min(order_purchase_timestamp)as first_order_time,
max(order_purchase_timestamp)as last_order_time
from Target.orders;
```

Query results Save results

Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	first_order_time	last_order_time	
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	

- Insights

Orders between first order date and last order date indicating that the data spans a meaningful time period for analysing order trends

3. Count the Cities & States of customers who ordered during the given period.

```
select
count(distinct geolocation_city)as no_of_cities,
count(distinct geolocation_state)as no_of_states
from Target.geolocation;
```

Query results

Job information	Results	Visualization	JSON
Row	no_of_cities	no_of_states	
1	8011	27	

- Insights

The customer base spans multiple cities across several states indicating a geographically diverse dataset suitable for regional analysis

2. In-depth Exploration:

1. Is there a growing trend in the no. of orders placed over the past years?

```
select
  extract(year from order_purchase_timestamp) as year,
  count(order_id) as No_Of_Orders
from Target.orders
group by extract(year from order_purchase_timestamp)
order by No_Of_Orders;
```

Query results

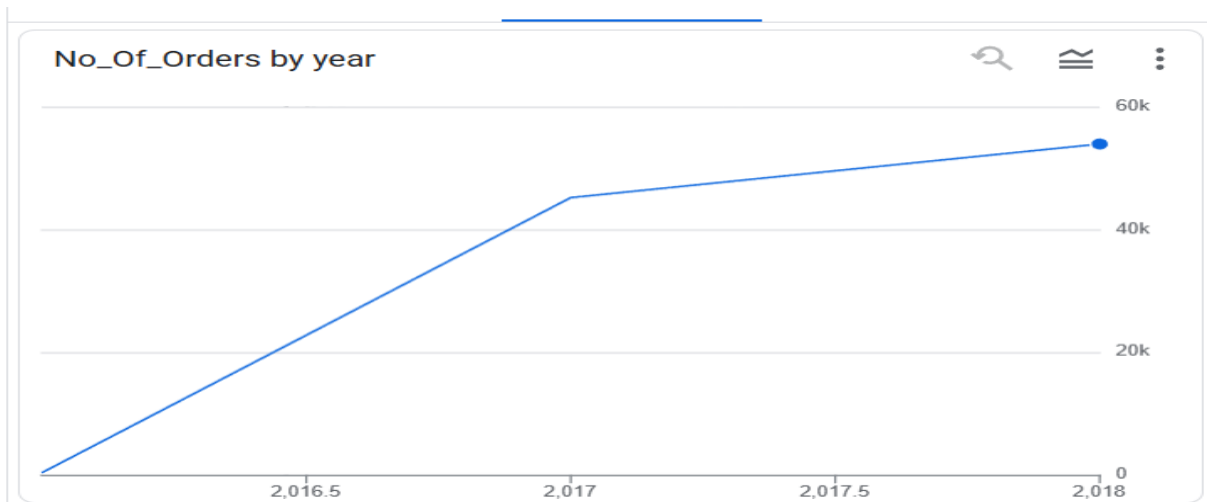
Job information		Results	Visualization	JSON	Execution details
Row	year ▼	No_Of_Orders ▼			
1	2016	329			
2	2017	45101			
3	2018	54011			

- Insights

The number of orders shows a strong year-on-year growth from 2016 to 2018.

Order volume increased significantly in 2017 and continued to grow in 2018,

indicating rapid platform adoption and sustained customer demand over time.



2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```

Select
extract(month from order_purchase_timestamp) as month,
count(order_id)as No_Of_Orders
from Target.orders
group by extract(month from order_purchase_timestamp)
order by month;

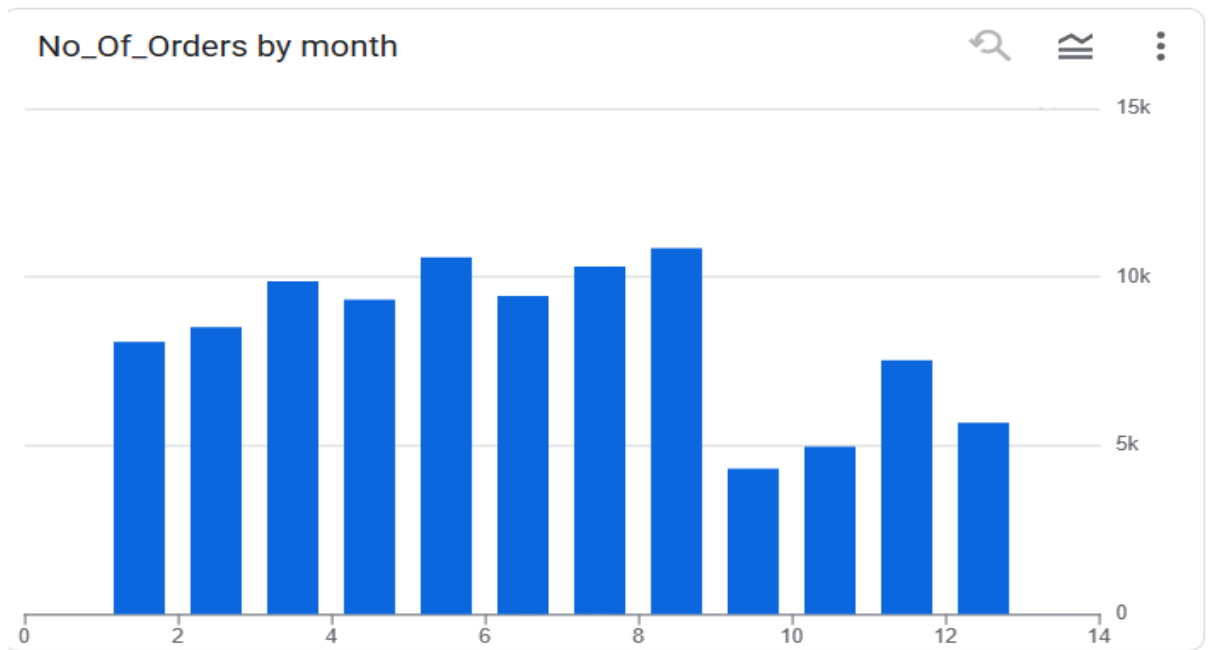
```

Query results				
Job information		Results	Visualization	JSON
Row	month ▼	No_Of_Orders ▼		
1	1	8069		
2	2	8508		
3	3	9893		
4	4	9343		
5	5	10573		
6	6	9412		
7	7	10318		
8	8	10843		
9	9	4305		
10	10	4959		

- Insights**

The number of orders varies across different months, indicating the presence of monthly seasonality.

Certain months consistently record higher order volumes, suggesting predictable demand patterns that can be leveraged for planning promotions, inventory, and logistics.



3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

```

with ab as
(
select order_id,
case when extract(hour from order_purchase_timestamp) between
0 and 6 then 'Dawn'
when extract(hour from order_purchase_timestamp) between 7 and
12 then 'Mornings'
when extract(hour from order_purchase_timestamp) between 13
and 18 then 'Afternoon'
else 'Night'
end as Time_Of_Day
from Target.orders
)
select Time_Of_Day,
count(order_id)as no_of_orders
from ab
group by Time_Of_Day
order by no_of_orders;

```

Query results

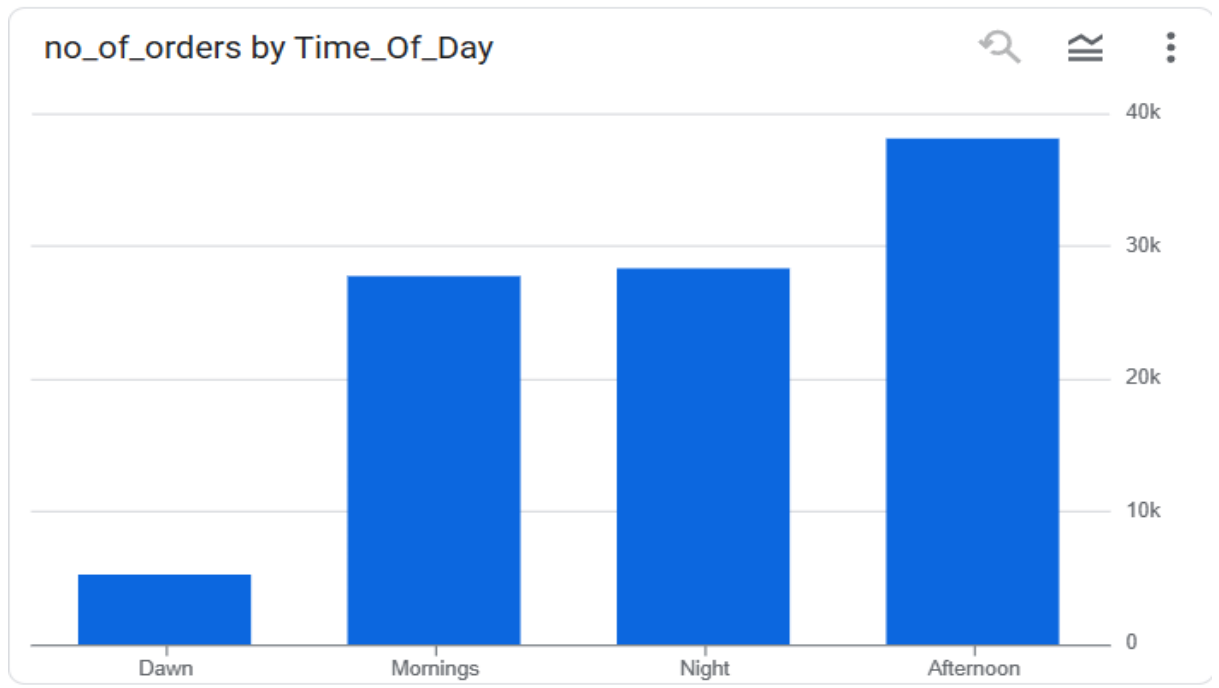
Job information		Results	Visualization	JSON	Execution details
Row	Time_Of_Day ▼	no_of_orders ▼			
1	Dawn	5242			
2	Mornings	27733			
3	Night	28331			
4	Afternoon	38135			

- Insights

Customer orders are predominantly placed during the Afternoon and Night hours,

indicating peak purchasing activity in the later part of the day.

Morning hours also show significant activity, while Dawn records the lowest order volume,suggesting limited customer engagement during early hours.



3. Evolution of E-commerce orders in the Brazil region:

1. Get the month on month no. of orders placed in each state.


```

select
c.customer_state,
extract(year from o.order_purchase_timestamp)as
year,
extract(month from o.order_purchase_timestamp)as
month,
count(o.order_id)as no_of_orders
from Target.customers c
join Target.orders o
on c.customer_id=o.customer_id
group by customer_state,
extract(year from o.order_purchase_timestamp),
extract(month from o.order_purchase_timestamp)
order by year,month,customer_state;

```

Query results

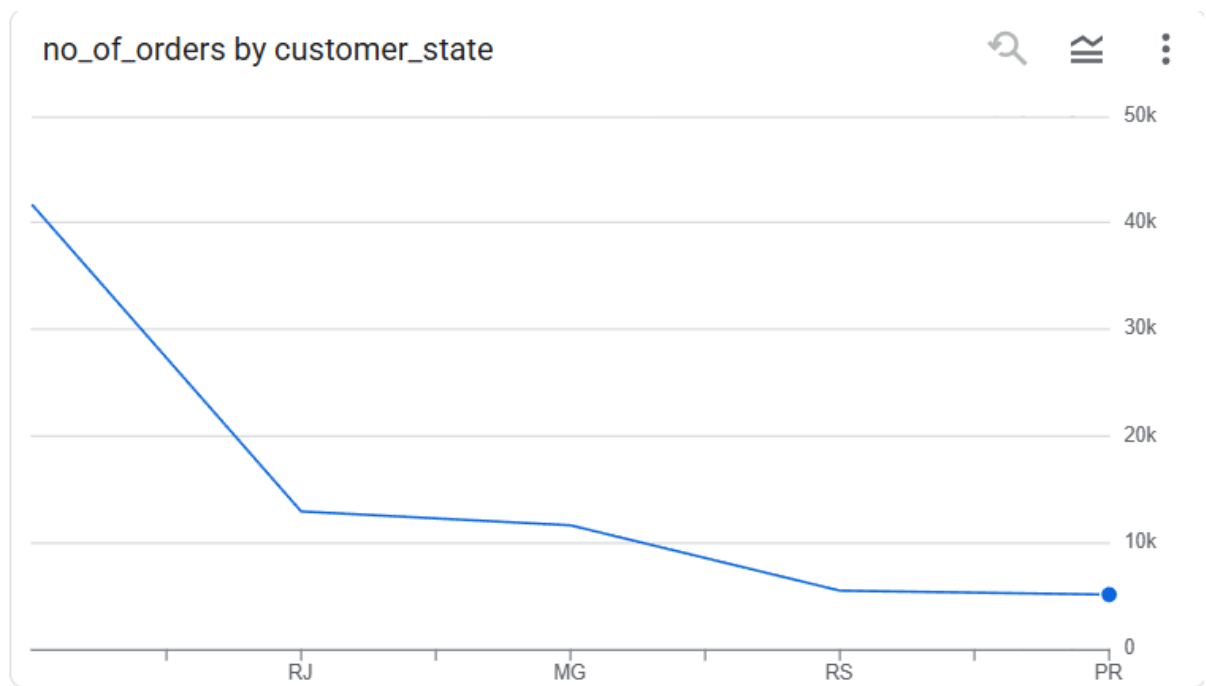
 Save re

Job information						Results	Visualization	JSON	Execution details	Execution g
Row	//	customer_state ▼	//	year ▼	//	month ▼	//	no_of_orders ▼	//	
1		RR		2016		9		1		
2		RS		2016		9		1		
3		SP		2016		9		2		
4		AL		2016		10		2		
5		BA		2016		10		4		
6		CE		2016		10		8		
7		DF		2016		10		6		
8		ES		2016		10		4		
9		GO		2016		10		9		
10		MA		2016		10		4		

- Insights

Month-on-month order volumes vary significantly across states, with a few states consistently contributing the majority of orders, indicating regional concentration of e-commerce activity in Brazil.

Orders with missing state information were excluded from this analysis to ensure accurate and meaningful state-level comparisons.



Only the top five states are visualized to avoid overcrowding and to emphasize key regional demand trends.

2. How are the customers distributed across all the states?

```

select
customer_state,
count(distinct customer_unique_id)as no_of_customers
from Target.customers
group by customer_state
order by no_of_customers desc;

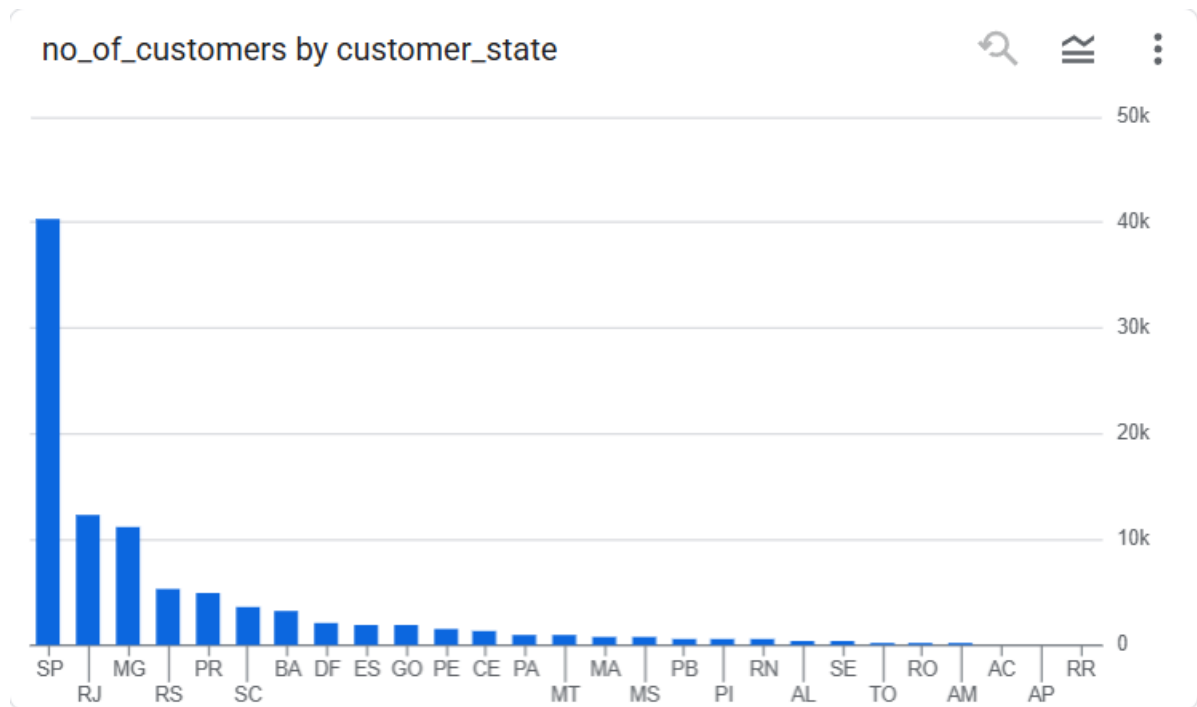
```

Query results

Job information		Results	Visualization	JSON	Execution details
Row	customer_state ▼	no_of_customers ▼			
1	SP	40302			
2	RJ	12384			
3	MG	11259			
4	RS	5277			
5	PR	4882			
6	SC	3534			
7	BA	3277			
8	DF	2075			
9	ES	1964			
10	GO	1952			

- Insights

Customers are unevenly distributed across states, with a small number of states accounting for a large share of the total customer base, indicating strong regional concentration of demand.



4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only). You can use the "payment_value" column in the payments table to get the cost of orders.

```

with ab as
(select extract(year from order_purchase_timestamp) as year,
sum(payment_value)as total_payment
from Target.orders o
join Target.payments p
using(order_id)
where extract(year from order_purchase_timestamp) in
(2017,2018)
and extract(month from order_purchase_timestamp)between 1 and
8
group by extract(year from order_purchase_timestamp))

select
round(
(max(CASE WHEN year = 2018 THEN total_payment END)
- max(CASE WHEN year = 2017 THEN total_payment END))
/ max(CASE WHEN year = 2017 THEN total_payment END) * 100,
2
) AS percent_increase
from ab;

```

Query results

Job information		Results	Visualization
Row	percent_increase		
1	136.98		

- Insights**
 The total cost of orders increased from 2017 to 2018 (Jan–Aug period), indicating growth in transaction value and increased customer spending over time.

2. Calculate the Total & Average value of order price for each state.

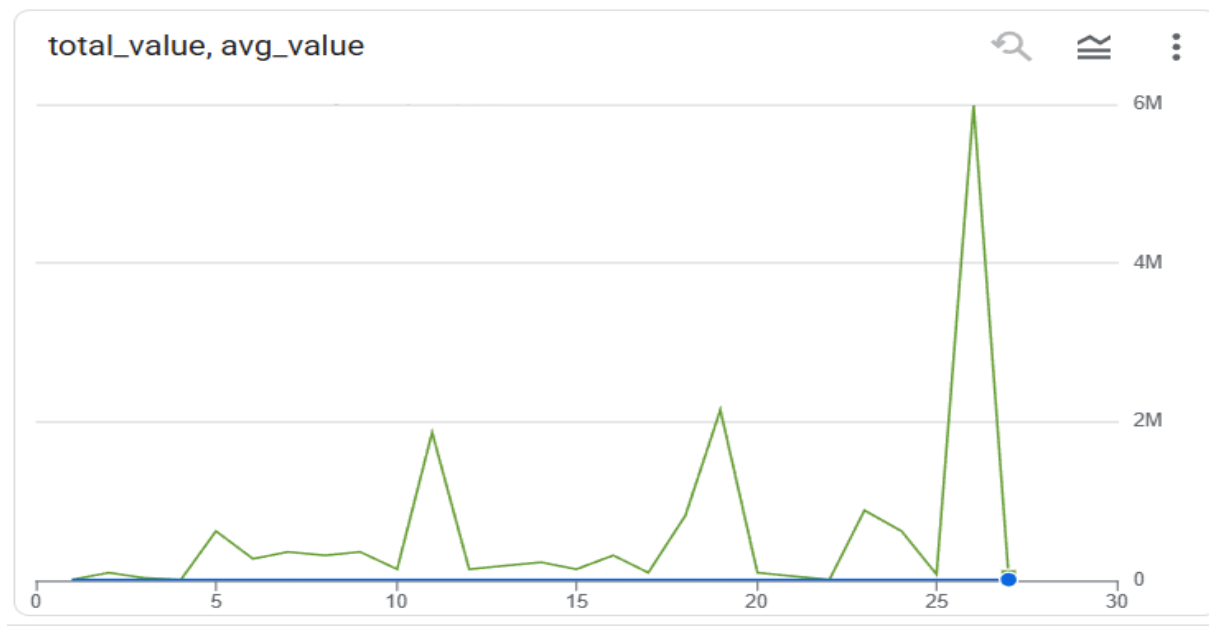
```
select
customer_state,
round(sum(payment_value),2)as total_value,
round(sum(payment_value)/count(distinct order_id),2)as
avg_value
from Target.customers c
join Target.orders o
using(customer_id)
join Target.payments p
using(order_id)
group by customer_state
order by customer_state;
```

Query results

Job information		Results	Visualization	JSON	Execution details
Row	customer_state ▼	total_value ▼	avg_value ▼		
1	AC	19680.62	242.97		
2	AL	96962.06	234.77		
3	AM	27966.93	188.97		
4	AP	16262.8	239.16		
5	BA	616645.82	182.44		
6	CE	279464.03	209.18		
7	DF	355141.08	165.95		
8	ES	325967.55	160.34		
9	GO	350092.31	173.31		
10	MA	152523.02	204.18		

- Insights

Some states generate high total revenue primarily due to large order volumes, while others show relatively higher average order values, indicating differences between volume-driven and value-driven markets.



3. Calculate the Total & Average value of order freight for each state.

```

select
customer_state,
round(sum(freight_value),2)as total_value,
round(sum(freight_value)/count(distinct order_id),2)as
avg_value
from Target.customers
left join Target.orders
using(customer_id)
join Target.order_items
using(order_id)
group by customer_state
order by customer_state;

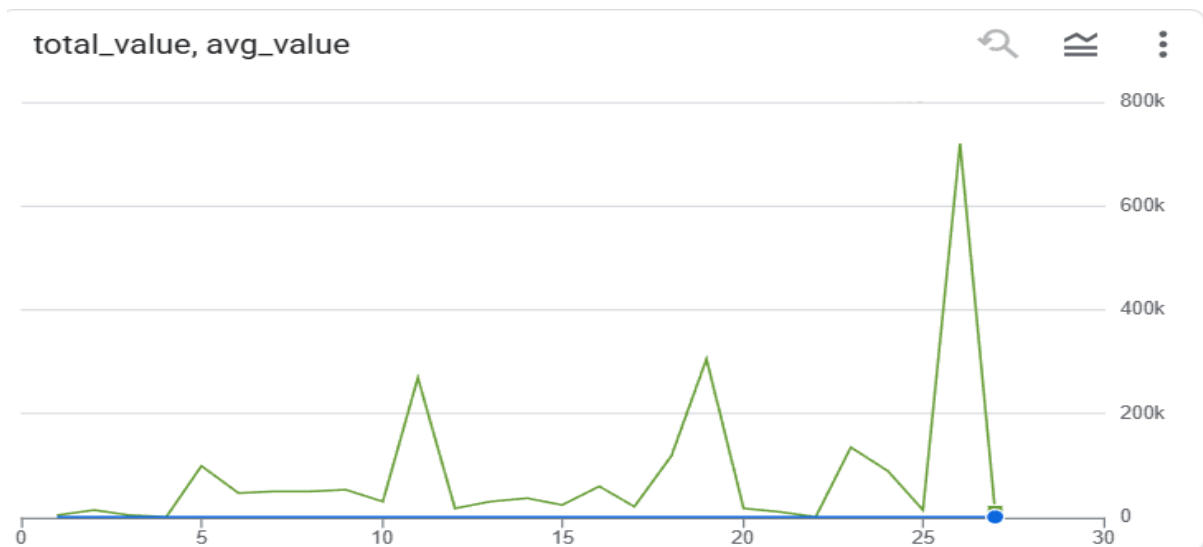
```

Query results

Job information		Results	Visualization	JSON	Execution details
Row	customer_state ▼	total_value ▼	avg_value ▼		
1	AC	3686.75	45.52		
2	AL	15914.59	38.72		
3	AM	5478.89	37.27		
4	AP	2788.5	41.01		
5	BA	100156.68	29.83		
6	CE	48351.59	36.44		
7	DF	50625.5	23.82		
8	ES	49764.6	24.58		
9	GO	53114.98	26.46		
10	MA	31523.77	42.6		

- Insights

Some states such as AC, AP, and MA have lower total freight value but higher average freight cost per order, indicating fewer orders but comparatively higher shipping costs, possibly due to distance or limited logistics infrastructure.



5. Analysis based on sales, freight and delivery time.

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- **time_to_deliver** = order_delivered_customer_date - order_purchase_timestamp
- **diff_estimated_delivery** = order_delivered_customer_date - order_estimated_delivery_date

```

select
order_id,
extract(day from(extract(date from
order_delivered_customer_date)-
(extract(date from order_purchase_timestamp))))as
time_to_deliver,
extract(day from(extract(date from
order_delivered_customer_date)-
(extract(date from
order_estimated_delivery_date))))as
diff_estimated_delivery
from Target.orders
where order_delivered_customer_date is not null
order by order_id;

```

Query results

Job information		Results	Visualization	JSON	Execution details
Row	order_id	time_to_deliver	diff_estimated_d...		
1	00010242fe8c5a6d1ba2dd792...	7	-9		
2	00018f77f2f0320c557190d7a1...	16	-3		
3	000229ec398224ef6ca0657da...	8	-14		
4	00024acbcd0a6daa1e931b038...	6	-6		
5	00042b26cf59d7ce69dfabb4e5...	25	-16		
6	00048cc3ae777c65dbb7d2a06...	7	-15		
7	00054e8431b9d7675808bcb81...	8	-17		
8	000576fe39319847cbb9d288c...	5	-16		
9	0005a1a1728c9d785b8e2b08b...	10	0		
10	0005f50442cb953dcd1d21e1fb...	2	-19		

- Insights

A significant number of orders show negative differences between actual and estimated delivery dates, indicating that many orders were delivered earlier than promised.

2. Find out the top 5 states with the highest & lowest average freight value.

```
(select
c.customer_state,
round(sum(freight_value)/count(distinct order_id),2)as
avg_freight_value,
'Highest Avg Freight'as category
from Target.customers c
join Target.orders o
using(customer_id)
join Target.order_items oi
using(order_id)
group by c.customer_state
order by avg_freight_value desc
limit 5)
```

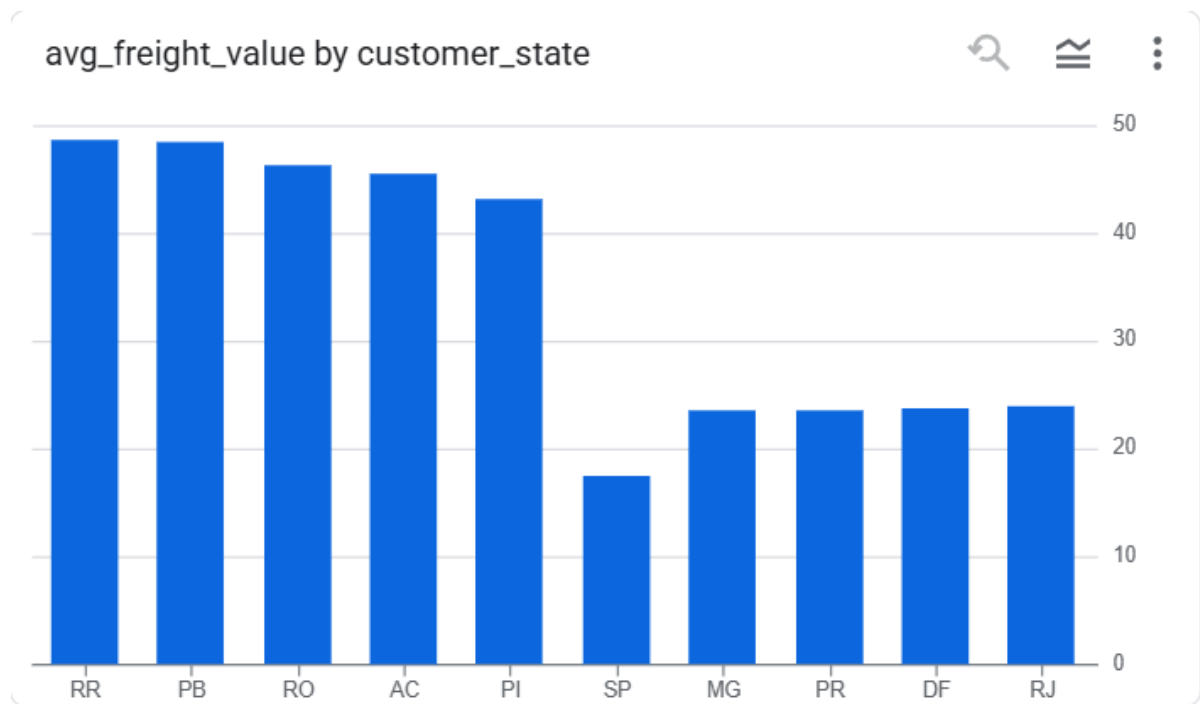
```
union all
```

```
(select
c.customer_state,
round(sum(freight_value)/count(distinct order_id),2)as
avg_freight_value,
'Lowest Avg Freight'as category
from Target.customers c
join Target.orders o
using(customer_id)
join Target.order_items oi
using(order_id)
group by c.customer_state
order by avg_freight_value
limit 5);
```

Query results

Job information	Results	Visualization	JSON	Execution details
Row	customer_state ▾	avg_freight_value ▾	category ▾	
1	RR	48.59	Highest Avg Freight	
2	PB	48.35	Highest Avg Freight	
3	RO	46.22	Highest Avg Freight	
4	AC	45.52	Highest Avg Freight	
5	PI	43.04	Highest Avg Freight	
6	SP	17.37	Lowest Avg Freight	
7	MG	23.46	Lowest Avg Freight	
8	PR	23.58	Lowest Avg Freight	
9	DF	23.82	Lowest Avg Freight	
10	RJ	23.95	Lowest Avg Freight	

- Insights**
 Average freight cost varies significantly across states,indicating differences in distance, logistics complexity, and delivery infrastructure.



3. Find out the top 5 states with the highest & lowest average delivery time.

```
(select
c.customer_state,
round(avg(extract(day from (extract(date from
o.order_delivered_customer_date)-extract(date from
o.order_purchase_timestamp))))),2)as Delivery_Time_Days,
'Highest Avg Delivery Time'as category
from Target.customers c
join Target.orders o
using(customer_id)
where o.order_delivered_customer_date is not null
group by c.customer_state
order by Delivery_Time_Days desc
limit 5)
```

union all

```
(select
c.customer_state,
round(avg(extract (day from (extract(date from
o.order_delivered_customer_date)-extract(date from
o.order_purchase_timestamp))))),2)as Delivery_Time_Days,
'Lowest Avg Delivery Time'as category
from Target.customers c
join Target.orders o
using(customer_id)
where o.order_delivered_customer_date is not null
group by c.customer_state
order by Delivery_Time_Days
limit 5)
```

Query results

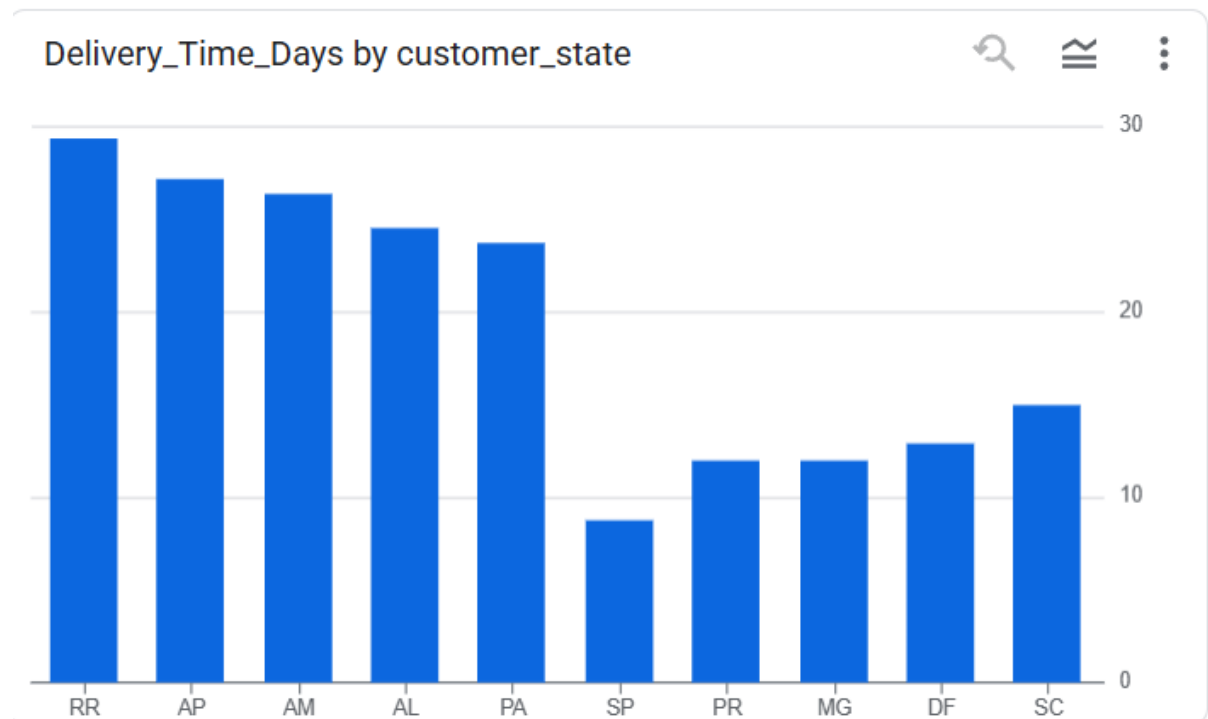
Job information		Results	Visualization	JSON	Execution details
Row	customer_state ▼	Delivery_Time_Da...	category ▼		
1	RR	29.34	Highest Avg Delivery Time		
2	AP	27.18	Highest Avg Delivery Time		
3	AM	26.36	Highest Avg Delivery Time		
4	AL	24.5	Highest Avg Delivery Time		
5	PA	23.73	Highest Avg Delivery Time		
6	SP	8.7	Lowest Avg Delivery Time		
7	PR	11.94	Lowest Avg Delivery Time		
8	MG	11.95	Lowest Avg Delivery Time		
9	DF	12.9	Lowest Avg Delivery Time		
10	SC	14.91	Lowest Avg Delivery Time		

- Insights

Average delivery times vary significantly across states.

A few states consistently experience longer delivery durations, while others benefit from faster order fulfillment.

This indicates regional differences in logistics efficiency and delivery infrastructure.



4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery. You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

```

with ab as
(select
c.customer_state,
round(avg(extract(day from(extract(date from
order_delivered_customer_date)-extract(date from
order_estimated_delivery_date)))),2)as Avg_Delivery_Time
from Target.customers c
join Target.orders o
using(customer_id)
group by c.customer_state)

select
*
from ab
where Avg_Delivery_Time<0
order by Avg_Delivery_Time
limit 5;

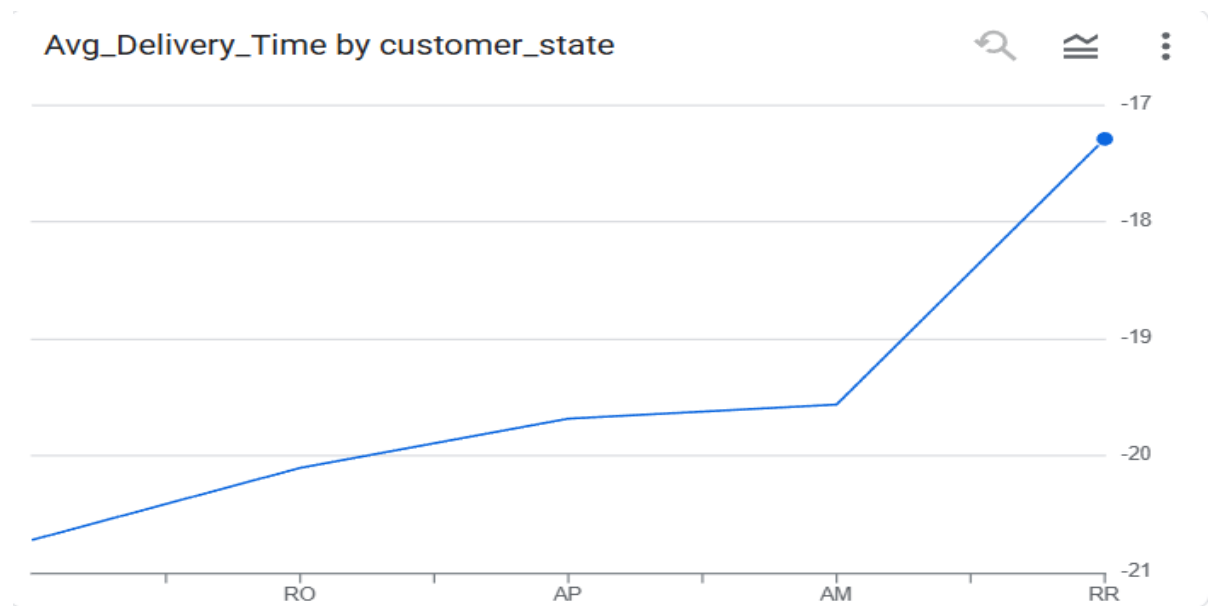
```

Query results

Job information		Results	Visualization	JSON
Row	customer_state	Avg_Delivery_Time		
1	AC	-20.72		
2	RO	-20.1		
3	AP	-19.69		
4	AM	-19.57		
5	RR	-17.29		

- Insights

These top five states indicate that orders are consistently delivered earlier than the estimated delivery date with orders arriving nearly 17-21 days ahead of schedule, reflecting strong logistics efficiency.



6. Analysis based on the payments:

1. Find the month on month no. of orders placed using different payment types.

```

select
extract(year from o.order_purchase_timestamp)as year,
extract(month from o.order_purchase_timestamp)as Month,
payment_type,
count(order_id)as no_of_orders
from Target.orders o
join Target.payments p
using(order_id)
group by extract(year from o.order_purchase_timestamp),
extract(month from o.order_purchase_timestamp),
payment_type
order by year,Month,payment_type;

```

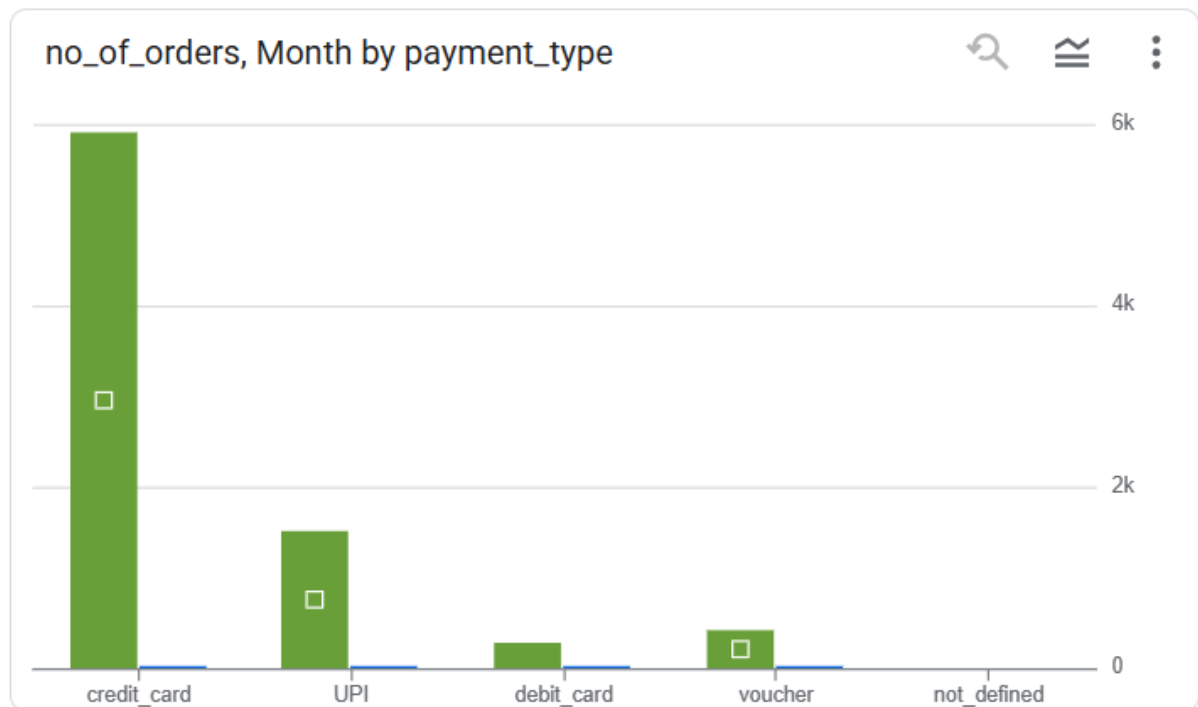
Query results



Job information	Results	Visualization	JSON	Execution details	Exec
Row	year	Month	payment_type	no_of_orders	
1	2016	9	credit_card	3	
2	2016	10	UPI	63	
3	2016	10	credit_card	254	
4	2016	10	debit_card	2	
5	2016	10	voucher	23	
6	2016	12	credit_card	1	
7	2017	1	UPI	197	
8	2017	1	credit_card	583	
9	2017	1	debit_card	9	
10	2017	1	voucher	61	

- Insights

Credit card is the most frequently used payment method across all months, consistently accounting for the highest number of orders.



2. Find the no. of orders placed on the basis of the payment installments that have been paid.

```

select
payment_installments,
count(distinct order_id)as no_of_orders
from Target.payments
group by payment_installments
order by payment_installments;

```

Query results

Job information		Results	Visualization	
Row	payment_installm...	no_of_orders ▼		
1	0	2		
2	1	49060		
3	2	12389		
4	3	10443		
5	4	7088		
6	5	5234		
7	6	3916		
8	7	1623		
9	8	4253		
10	9	644		

- Insights

Most customers prefer paying in a single installment, with installment-based payments declining sharply as the number of installments increases, indicating limited adoption of long-term EMI options.

